Graph Transformations: Single Transformations

Questions 1–70

Question 1. Translations Describe the transformation that maps $y = \mathrm{f}(x)$ onto each of these:

a) $y = \mathrm{f}(x - 6)$
b) $y = \mathrm{f}(x) + 8$
c) $y = \mathrm{f}(x + 4)$
d) $y = \mathrm{f}(x) - 11$

Question 2. Describe the transformation that maps $y = x^2$ onto each of these:

a) $y = (x - 9)^2$
b) $y = (x + 5)^2$
c) $y = x^2 + 2$
d) $y = x^2 - 10$
e) $y + 4 = x^2$
f) $y - 8 = x^2$
g) $y = (x + 7)^2 - 3$
h) $y = (x - 1)^2 + 7$

Question 3. Describe the transformation that maps $y = x^3$ onto each of these:

a) $y = (x + 4)^3$
b) $y = (x - 2)^3$
c) $y = x^3 - 7$
d) $y = x^3 + 15$
e) $y + 9 = x^3$
f) $y = (x - 3)^3 + 8$

Question 4. Write down the image of $y = 9x + 2$ after a translation by the vector $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$

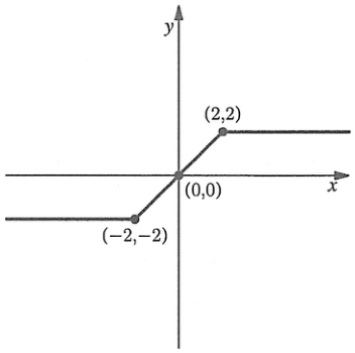
Question 5. The following functions are to be translated by the vector $\begin{pmatrix} 4 \\ -7 \end{pmatrix}$. Write down the image in each case.

a) $y = 4 - 6x$
b) $y = x^2$
c) $y = x^2 + 3$
d) $y = (x + 2)(x - 8)$
e) $y = (x - 9)(x + 1)$
f) $y = 3^x$
g) $y = e^x - 5$
h) $y = \frac{1}{x}$
i) $y = \frac{1}{x^2}$
j) $y = \sqrt{x}$
k) $y = \cos x$

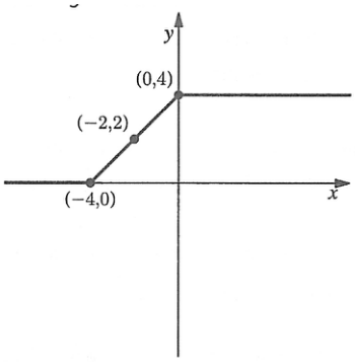
Question 6. The point $P(7, -5)$ lies on the graph with equation $y = \mathrm{f}(x)$. State the coordinates of the image of P after $y = \mathrm{f}(x)$ is mapped to each of these:

a) $y = \mathrm{f}(x - 8)$
b) $y = \mathrm{f}(x) + 3$
c) $y = \mathrm{f}(x + 12)$
d) $y = \mathrm{f}(x + 1) - 5$

Question 7. The graph of $y = \mathrm{f}(x)$ is shown in the diagram below.



Write down the equation of the graph shown in the diagram below.



Question 8. Describe the transformation that maps $y = 3x^4$ onto the image $y = 3(x + 4)^4 - 10$

Question 9. Describe the transformation that maps $y = x^8 + 7$ onto the image $y = (x - 8)^8$

Question 10. Describe the transformation that maps $y = \sqrt{x} + 2$ onto the image $y = \sqrt{x + 2}$

Question 11. Prerequisite knowledge for Q11–Q13: Logarithms Find the translation vector that maps $y = \log_5 x$ onto the image $y = \log_5\left(\frac{x-4}{25}\right)$.

Question 12. Find the translation vector that maps $y = \log_3 x$ onto the image $y = \log_3(9(x-2))$.

Question 13. Fully describe the transformation that maps $y = \ln x$ onto the image $y = \ln\left(\frac{x+2e}{e^5}\right)$.

Question 14. Find the image of $x^2 + y^2 = 9$ after a translation by the vector $\begin{pmatrix} -7 \\ 5 \end{pmatrix}$.

Question 15. Find the image of $x^2 + y^2 = 3$ after a translation by the vector $\begin{pmatrix} 2 \\ -8 \end{pmatrix}$.

Question 16. Find the image of $(x-2)^2 + y^2 = 10$ after a translation by the vector $\begin{pmatrix} -9 \\ -3 \end{pmatrix}$.

Question 17. Find the image of $(x+10)^2+(y-3)^2=16$ after a translation by the vector $\begin{pmatrix}2\\-1\end{pmatrix}$.

Question 18. Find the image of $xy+x^2-y=10$ after a translation by the vector $\begin{pmatrix}-3\\4\end{pmatrix}$.

Question 19. Find the image of $y = x^3 - 2x^2 + 8x + 2$ after a translation by the vector $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$.

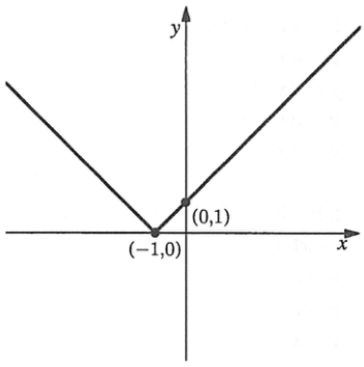
Question 20. Find the image of $y = 8x^4 - 4x^8$ after a translation by the vector $\begin{pmatrix} 4 \\ -8 \end{pmatrix}$.

Question 21. Find the image of $y = \frac{3}{x^2}$ after a translation by the vector $\begin{pmatrix} -2 \\ -6 \end{pmatrix}$.

Question 22. Find the image of $y = -\frac{4}{x-7}$ after a translation by the vector $\begin{pmatrix} 7 \\ 0 \end{pmatrix}$.

Question 23. Fully describe the transformation that maps $y = 4 \times 4^x$ onto the image $y = 4^{x-5}$.

Question 24. The diagram below shows the graph of $y = f(x)$. It is given that $f(x)$ is symmetric about $x = -1$.



On separate diagrams, draw a sketch of each of these:

a) $y = f(x + 1)$
b) $y = f(x) - 2$
c) $y = f(x - 4) + 1$
d) $y - 2 = f(x + 2)$

Question 25. Stretches Describe the transformation that maps $y = f(x)$ onto each of these:

a) $y = f(5x)$
b) $y = \frac{1}{6}f(x)$
c) $y = f(\frac{2}{3}x)$

Question 26. Describe the transformation that maps $y = x^2 - 3$ onto each of these:

a) $y = 3x^2 - 9$
b) $y = 9x^2 - 3$

Question 27. Describe the transformation that maps $y = x^3 + 2$ onto each of these:

a) $y = 8x^3 + 2$
b) $y = 8x^3 + 16$
c) $y = \frac{1}{8}x^3 + \frac{1}{4}$
d) $y = \frac{1}{8}x^3 + 2$

Question 28. Write down the image of $y = 5x + 3$ after:

a) a stretch, parallel to the x-axis, scale factor $\frac{1}{4}$
b) a stretch, parallel to the y-axis, scale factor $\frac{1}{5}$

Question 29. The following functions are to be stretched, parallel to the x-axis, scale factor 3. Write down the image in each case.

a) $y = 4 - 6x$
b) $y = x^2$
c) $y = x^2 + 3$
d) $y = (x + 2)(x - 8)$
e) $y = (x - 9)(x + 1)$
f) $y = 3^x$
g) $y = e^x - 5$
h) $y = \frac{1}{x}$
i) $y = \frac{1}{x^2}$
j) $y = \sqrt{x}$
k) $y = \cos x$

Question 30. The following functions are to be stretched, parallel to the y-axis, scale factor 3. Write down the image in each case.

a) $y = 4 - 6x$
b) $y = x^2$
c) $y = x^2 + 3$
d) $y = (x + 2)(x - 8)$
e) $y = (x - 9)(x + 1)$
f) $y = 3^x$
g) $y = e^x - 5$
h) $y = \frac{1}{x}$
i) $y = \frac{1}{x^2}$
j) $y = \sqrt{x}$
k) $y = \cos x$

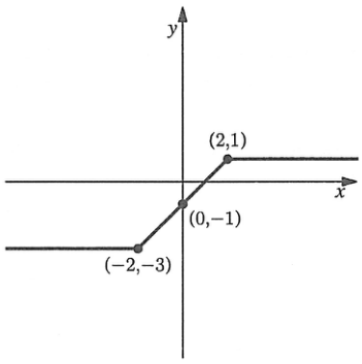
Question 31. The point $P(-2, 10)$ lies on the graph with equation $y = f(x)$. State the coordinates of the image of P after $y = f(x)$ is mapped to each of these:

a) $y = 6f(x)$
b) $y = f(\frac{1}{2}x)$
c) $y = \frac{2}{5}f(x)$
d) $y = f(6x)$

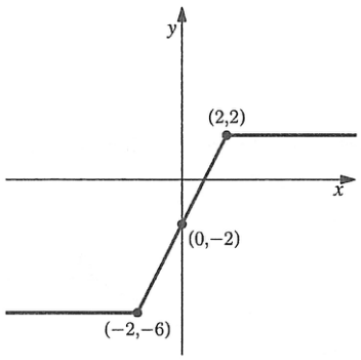
Question 32. Describe the transformation that maps $y = \sin x$ onto the image $y = \sin 5x$.

Question 33. Describe the transformation that maps $y = \cos 8x$ onto the image $y = 4\cos 8x$.

Question 34. The graph of $y = f(x)$ is shown in the diagram below.



Write down the equation of the graph shown in the diagram below.



Question 35. Describe the transformation that maps $y = (x + 3)^3$ onto the image $y = (\frac{1}{4}x + 3)^3$.

Question 36. Describe the transformation that maps $y = x^3$ onto the image $y = 27x^3$ as:

a) a stretch parallel to the y-axis,

b) a stretch parallel to the x-axis.

Question 37. Describe the transformation that maps $y = \frac{1}{x}$ onto the image $y = \frac{8}{x}$ as:

a) a stretch parallel to the y-axis,

b) a stretch parallel to the x-axis.

Question 38. Describe the transformation that maps $y = \frac{1}{x^2}$ onto the image $y = \frac{100}{x^2}$ as:

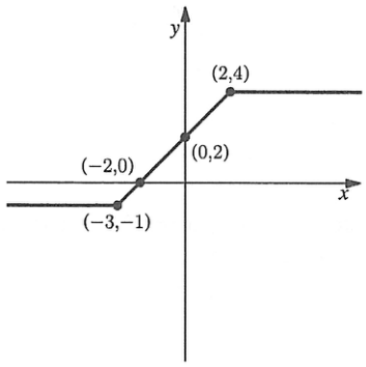
a) a stretch parallel to the y-axis,

b) a stretch parallel to the x-axis.

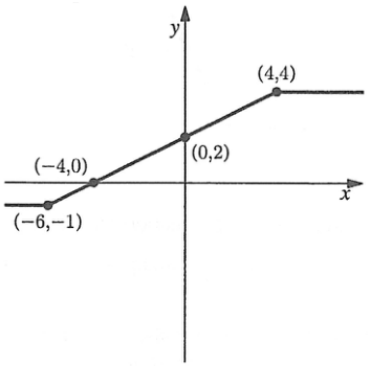
Question 39. Describe the transformation that maps $x^2 + y^2 = 16$ onto the image $x^2 + 9y^2 = 16$.

Question 40. Describe the transformation that maps $12x^2 + y^2 = 9$ onto the image $x^2 + y^2 = 9$.

Question 41. The graph of $y = f(x)$ is shown in the diagram below.



Write down the equation of the graph shown in the diagram below.

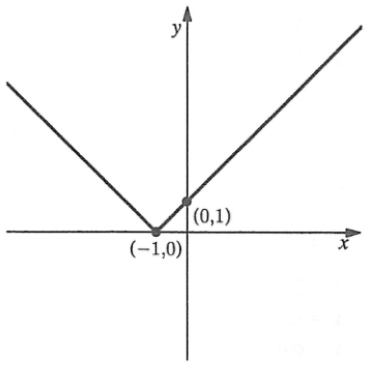


Question 42. Describe the single transformation that maps $y = 3(x-2)^2 + 5$ onto the image $y = 12(x-1)^2 + 5$.

Question 43. Describe the single transformation that maps $y = 4(x + 8)^2 - 2$ onto the image $y = 64(x + 2)^2 - 2$.

Question 44. Describe the single transformation that maps $y = 5(x - \frac{1}{4})^2 + 1$ onto the image $y = \frac{5}{64}(x - 2)^2 + 1$.

Question 45. The diagram below shows the graph of $y = f(x)$. It is given that $f(x)$ is symmetric about $x = -1$.



On separate diagrams, draw a sketch of each of these:

a) $y = f(2x)$
b) $y = 4f(x)$
c) $3y = f(x)$
d) $y = f(\frac{1}{3}x)$

Question 46. Reflections Describe the transformation that maps $y = f(x)$ onto each of these:

a) $y = f(-x)$
b) $y = -f(x)$

Question 47. Describe the transformation that maps $y = (x+3)^2 - 4$ onto each of these:

a) $y = 4 - (x+3)^2$

b) $y = (3-x)^2 - 4$

Question 48. Prerequisite knowledge for Q48: e^x **&** $\ln x$ Describe the transformation that maps $y = e^x - 2$ onto each of these:

a) $y = e^{-x} - 2$

b) $y = 2 - e^x$

c) $y = \frac{1-2e^x}{e^x}$

Question 49. Write down the image of $y = 8x - 9$ after:

a) a reflection in the x-axis,

b) a reflection in the y-axis.

Question 50. Write down the image of $y = 3 - 6x$ after:

a) a reflection in the x-axis,

b) a reflection in the y-axis.

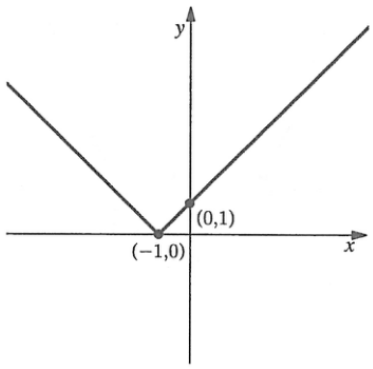
Question 51. The following functions are to be reflected in the y-axis. Identify which (if any) of these functions will remain unchanged:

a) $y = 5 - x$
b) $y = x^2 + 2$
c) $y = x^3$
d) $y = 4^x$
e) $y = \frac{1}{x}$
f) $y = \frac{1}{x^2}$
g) $y = \sin x$
h) $y = \cos x$
i) $y = \frac{e^x + e^{-x}}{2}$

Question 52. The following functions are to be reflected in the x-axis. Identify which (if any) of these functions will remain unchanged:

a) $y = 5 - x$
b) $y = x^2 + 2$
c) $y = x^3$
d) $y = 4^x$
e) $y = \frac{1}{x}$
f) $y = \frac{1}{x^2}$
g) $y = \sin x$
h) $y = \cos x$
i) $y = \frac{e^x + e^{-x}}{2}$

Question 53. The diagram below shows the graph of $y = f(x)$. It is given that $f(x)$ is symmetric about $x = -1$.



On separate diagrams, draw a sketch of each of these:

a) $y = f(-x)$

b) $y = -f(x)$

Question 54. The point $P(4, -1)$ lies on the graph with equation $y = f(x)$. State the coordinates of the image of P after $y = f(x)$ is mapped to each of these:

a) $y = -f(x)$

b) $y = f(-x)$

Question 55. Mix Describe the transformation that would map $y = \sqrt{x}$ onto each of these:

a) $y = \sqrt{x} - 3$
b) $y = \sqrt{2x}$
c) $y = \sqrt{x + 9}$
d) $y = -\sqrt{x}$
e) $y = 8\sqrt{x}$

Question 56. It is given that $f(x) = 2x^2$ and $g(x) = x - 1$. Show that $y = f(x) + 1$ and $y = g(3x)$ do not intersect.

Question 57. It is given that $f(x) = (x-1)^2 - 1$ and $g(x) = 3x$. Show that $y = 2f(x)$ and $y = g(x-2)$ intersect at two distinct points.

Question 58. Describe a single transformation that will map $y = \frac{16}{x+1}$ onto the image $y = \frac{9}{x+1}$.

Question 59. It is given that $f(x) = \frac{1}{x^2}$. State the asymptotes of the following graphs:

a) $y = f(x + 2)$
b) $y = -f(x)$
c) $y = f(x - 5) + 4$
d) $y = f(\frac{1}{3}x)$

Question 60. It is given that $y = f(x)$ is a polynomial of degree 3. It is also given that $f(-2) = 0$, $f(5) = 0$ and $f(7) = 0$. $y = f(x)$ is mapped to the image $y = g(x)$. It is now given that $g(a) = 0$, $g(b) = 0$ and $g(c) = 0$. State the values of a, b and c when:

a) $g(x) = f(x - 2)$
b) $g(x) = f(x + 10)$
c) $g(x) = 4f(x)$
d) $g(x) = f(2x)$
e) $g(x) = f(\frac{1}{3}x)$
f) $g(x) = f(-x)$

Question 61. It is given that $f(x) = (x-2)(x+4)(x+5)$.

a) Solve $f(x) = 0$
b) Solve $f(x-2) = 0$
c) Solve $f(2x) = 0$
d) Solve $f(\frac{1}{3}x) = 0$
e) Solve $f(-x) = 0$
f) Solve $f(x+15) = 0$

Question 62. It is given that $f(x) = \frac{8}{x-6}$. State the equation of the image of $f(x)$ after each of these transformations:

a) A translation by the vector $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$
b) A stretch, parallel to the y-axis, scale factor $\frac{1}{2}$
c) A reflection in the y-axis
d) A stretch, parallel to the x-axis, scale factor $\frac{1}{4}$

Question 63. It is given that $y = f(x)$ is a polynomial of degree 3. It is also given that $f(-3) = 0$, $f(6) = 0$ and $f(8) = 0$. $y = f(x)$ is mapped to the image $y = g(x)$, where $g(x) = f(x - k)$ and k is a constant. It is also given that $y = g(x)$ passes through $(1, 0)$. State the possible values of k.

Question 64. It is given that $y = f(x)$ is a polynomial of degree 3. It is also given that $f(-8) = 0$, $f(-2) = 0$ and $f(3) = 0$. $y = f(x)$ is mapped to the image $y = g(x)$, where $g(x) = f(kx)$, where k is a constant. It is also given that $y = g(x)$ passes through $(-1, 0)$. State the possible values of k.

Question 65. It is given that $f(x) = 4^x$. The graph of $y = f(x)$ is mapped onto the image $y = g(x)$, where $g(x) = 4^{x-3}$.

a) Fully describe the transformation using a translation.

b) Fully describe the transformation using a stretch.

Question 66. Fully describe the single transformation that maps $y = x^2$ onto the curve $y = x^2 + 8x - 3$.

Question 67. Fully describe the single transformation that maps $y = 2x^2$ onto the curve $y = 2x^2 - 8x - 5$.

Question 68. It is given that $f(x) = 3x^2 - 2x + 1$. Find the new equation of the curve in the form $y = ax^2 + bx + c$ after each of these transformations:

a) A reflection in the y-axis

b) A translation by the vector $\begin{pmatrix} 3 \\ -6 \end{pmatrix}$

c) A stretch, parallel to the x-axis, scale factor $\frac{1}{3}$

Question 69. It is given that $f(x) = 2x^4 - 7x^3 - 4x^2$.

a) Solve $f(x) = 0$

It is given that $g(x) = 2(x+3)^4 - 7(x+3)^2 - 4(x+3)^2$.

b) Use your answer to a) to solve $g(x) = 0$

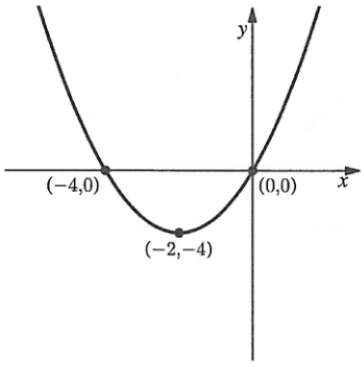
It is given that $h(x) = 162x^4 - 189x^3 - 36x^2$.

c) Use your answer to a) to solve $h(x) = 0$

It is given that $m(x) = 2x^4 + 7x^3 - 4x^2$.

d) Use your answer to a) to solve $m(x) = 0$

Question 70. The diagram below shows the graph of $y = f(x)$. It is given that $f(x)$ is symmetric about $x = -2$.



On separate diagrams, draw a sketch of each of these:

a) $y = -f(x)$

b) $y = f(x-3)$

c) $y = \frac{5}{2}f(x)$